Mingming Cai

206-790-3693 mingmingcai306@gmail.com Mingming's LinkedIn Mingming's Homepage

Biography

- Ph.D. candidate in sustainable transportation and human-centered AI, with a concurrent M.S. in Statistics from UW.
- Academic journey centered on applying Statistical, Machine Learning, and Deep Learning models to travel demand prediction | Industry experience with state-of-the-art Generative AI models
- Proficient in Python (5+ years), R (5+ years), and SQL (3+ years), with dedicated experience in software system design using C++ and C# (1+ years), distributed computing using MapReduce and Spark (2+ years), Git (2+ years), Linux (2+ year), and AWS (1+ year).
- Passionate about state-of-art and responsible AI techniques (Robustness, Fairness) to promote human-centered AI services.
- A collaborative person with effective communication and contributions to diverse projects within interdisciplinary teams.

Education

University of Washington September 2020 - May 2025 (Expected)
Ph.D. in Transportation Economics and Regional Planning (GPA: 3.9 / 4.0) - Minor: Data Science *Seattle, WA*
- Relevant Coursework: Machine Learning for Big Data (Recommendation System, NLP, Social-Network Graph Analysis), Statistical Modeling & Prediction & Computing (TensorFlow, PyTorch), Time Series Analysis, Statistical Analysis of Social Networks

University of Washington September 2023 - May 2025 (Expected)
M.S. in Statistics (GPA: 3.8 / 4.0) - Minor: Machine Learning *Seattle, WA*
- Relevant Coursework: Causal Modeling, Bayesian Statistics, Design and Analysis of Experiments

Wuhan University September 2017 - June 2020
M.S. in Land Resource Management (GPA: 3.9 / 4.0) *Wuhan, China*
- Relevant Coursework: Python Programming, Database System (SQL), Data Visualization (D3.js), Operational Research

Wuhan University September 2013 - June 2017
B.E. in Natural & Land Resources Management (GPA: 3.8 / 4.0) *Wuhan, China*
- Relevant Coursework: Advanced Mathematics, Linear Algebra, Probability Theory and Statistics, Database Principles (SQL, Oracle), Programming Language Design (C), Software System Design (C, C++), Microeconomics, Macroeconomics, Data Structures and Algorithms, Systems Programming (C, C++)

Research Projects

Predicting Ride-hailing Demand in Low-density Areas: A Probabilistic Multi-task GNN Approach | *Leading Researcher*
Funded by Washington State Department of Transportation, ongoing
- Developed probabilistic spatiotemporal GNNs to quantifies uncertainty in sparse travel demand generation, integrating probabilistic modeling with graph embedding.
- Showcased the enhanced predictive performance of probabilistic modeling incorporated GNNS for uncertainty quantification, surpassing baseline statistical, machine learning, and deep learning models (XGBoost, GCNs, GraphSAGE).

Airbnb Recommender System: NLP-Driven Semantic Information Mining for User Preferences | *Leading Researcher*
Funded by Paul G. Allen School of Computer Science & Engineering, University of Washington, 2023
- Fine-tuned BERT for end-to-end sentiment analysis and developed topic modeling pipelines (BERTopic) in PyTorch to extract semantic features from millions of user reviews, enhancing personalized recommendation systems and mitigating popularity bias and cold-start issues.
- Optimized large-scale text preprocessing workflows using NLP libraries to streamline data preparation for downstream modeling.
- Accelerated model training on large datasets by tuning hyperparameters through coherence analysis and leveraging distributed computing with Pyspark and Google Cloud.

Time-series Forecasting of Real Estate Taxes using Deep Learning Techniques | *Leading Researcher*
Funded by National Natural Science Foundation of China, 2017
- Built time-series forecasting models (ARIMA, exponential smoothing, RNNs) to predict real estate tax revenue for Shanghai pilot projects, achieving 72.9% accuracy with RNNs and demonstrating 5%–12% lower accuracy with traditional models.
- Optimized the data pipeline and engineered predictive features through variable transformation, missing data imputation, and input standardization to improve model robustness.
- Led the research and published findings in a peer-reviewed academic journal.

Work Experience

Data Scientist Intern September 2024 - December 2024
Amazon *Sunnyvale, CA*

- Identified research problems and developed scientific solutions to boost RAG system performance, enhancing Amazon LLM Q&A accuracy to user queries.
- Improved LLM inference by adapting and advancing state-of-the-art information retrieval methods, raising Recall@5/10 for context retrieval and cutting token latency by over 50%.
- Implemented methodological innovations in Python, independently tested prototypes, and collaborated with software engineers to integrate solutions into production.

Data Analyst - Modeling Intern June 2023 - August 2023
Chicago Metropolitan Agency for Planning *Chicago, IL*

- Optimized EMME auto-routing algorithm for simulation workflows and implemented cross-validation against Google Maps routing, reducing highly mismatched routes by 70% and moderately mismatched routes by 45%.
- Developed and deployed interactive online maps of route validation metrics by leveraging web-based mapping technologies.
- Engineered Python scripts to streamline the measurement of transit accessibility and transportation greenhouse gas emissions, integrating multi-source data inputs.

Research Scientist Intern - ML Modeling June 2020 - August 2020
Wuhan Transportation Development Strategy Institute *Wuhan, China*

- Implemented Conv-LSTM neural networks to predict citywide dockless bike sharing demand, capturing spatio-temporal dependencies and improving prediction accuracy (MAPE) by 5%-15% compared to ARIMA, LSTM, GRU.
- Developed performance-optimized stacked ensemble models (XGBoost, LightGBM, CatBoost) to predict ride-hailing travel time using city-scale mobility data, achieving a MAPE of 3%-18% for off-peak and peak periods.
- Engineered web scraping pipelines in Python (Beautiful Soup, Scrapy) to collect real-time open data on transit services for downstream modeling and analysis.

Data Science and Big Data Analytics Intern June 2018 - August 2018
Wuhan Geomatics Institute *Wuhan, China*

- Developed spatio-temporal data mining algorithms to identify short-term and long-term citywide commute behavior patterns from cell phone signaling data, supporting government decision-making on transportation infrastructure investments.
- Enhanced data visualization pipelines using Vega and D3.js, delivering insights published in the 2019 Wuhan Municipal Government annual report.

Certificates & Awards

- 2024 Outstanding PhD Student Award for Academic Achievement and Leadership, University of Washington
- 2022 Special Student Service and Contribution, University of Washington
- 2019 Grand Prize, Geographic Information Science and Technology Innovation Contest, Ministry of Natural Resources of China
- 2018 First-level Scholarships, Wuhan University (Top 5 %)
- 2017 Outstanding Undergraduate Thesis, Department of Education of Hubei Province, China
- 2016 Honorable Mention, Interdisciplinary Contest in Modeling , Consortium for Mathematics and its Applications (COMAP)
- 2015 National Scholarship, Ministry of Education of China (Top 1 %)
- 2014 National Scholarship, Ministry of Education of China (Top 1 %)

Technical Skills

Programming Languages: Python, R, SQL, JavaScript, Html
Python Libraries for ML/DL: PyTorch, TensorFlow, Keras, Scikit-learn, SciPy, Numpy, Pandas, Geopandas
NLP: Scikit-learn, Genism, spaCy, Pyspark MLlib
Statistical Analysis: Time Series Analysis, Causal Inference, Bayesian Inference, Network Analysis, Experiment Design and Analysis
Distributed Computing: Spark, Pyspark, MapReduce, Hive, Yarn
Spatial Analysis & Simulation: ArcGIS, QGIS, GeoDa, SUMO, MATSim
Data Visualization: Tableau, Power BI, Vega, D3.js, CSS